

Structural Analysis & Review Report

Chemical anchors, post-installed dowels and reinforcement in concrete
Assessment against AS 5216 and AS 3600

INCOMPLETE

Required data is missing, so adequacy cannot be confirmed.

Project	Demo Warehouse Extension (fictitious)
Reference	DEMO-001
Client	Demo Client Pty Ltd
Drawings reviewed	sample_structural_drawing.pdf
Report date	06 October 2026
Revision	A
Prepared by	(software-assisted draft)
Checked by	(pending)
Governing standard	AS 5216:2021

This report was prepared with software assistance from the drawings supplied. Extracted data, product specifications and calculation results are traceable to their sources in the sections that follow. It is a design-review aid and does not replace the judgement of the responsible registered/chartered structural engineer, who must verify the inputs, product data and results before reliance.

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1. Executive summary

Outcome: INCOMPLETE. Required data is missing, so adequacy cannot be confirmed.

Case	Description	Adhesive / data source	Governing check	Max util.	Status
A1	M20 rods, hef 200 mm 2x2 anchors	HIT-RE 500 V3 attached documents	Combined tension + shear - concrete failure modes [Drawing]	0.81	PASS
R1	N16 dowels, hef 400 mm	HIT-RE 500 V3 attached documents	Bar tension capacity - development length (AS 3600 13.1.2 / AS 5216...	0.19	INCOMPLETE

Key findings

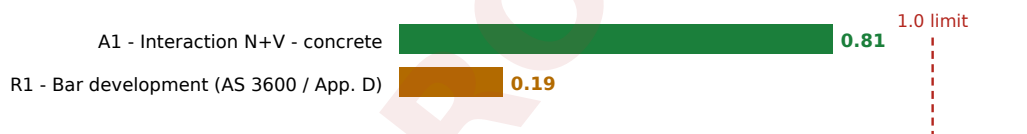
MAJOR Information not on the drawing (RFI) - M20 chemical anchor (p.1): direction of the shear force is not stated - assumed to act towards the nearest free edge (conservative). State the shear direction and whether it can reverse.

MAJOR Information not on the drawing (RFI) - N16 dowel (p.2): reinforcement grade not stated - D500N assumed (AS/NZS 4671).

MAJOR R1: adequacy cannot be confirmed - No f_{bd} (EAD 330087) value found for HIT-RE 500 V3: assumed $\geq$ the EAD table value (3.2 MPa). Confirm against the product's ETA-Rebar. HIT-RE 500 V3 is not confirmed as qualified for post-installed rebar (EAD 330087 / TR 023) -...

MAJOR R1: product gap - Post-installed rebar qualification - Required: EAD 330087 / ETA-Rebar with f_{bd} . Product documents: not confirmed. Rebar-as-anchor design (EAD 330499) is an alternative if bond data exist.

Governing utilisation by case



2. Scope, documents and method

2.1 Documents reviewed

File	Pages	Text layer	Reading method
sample_structural_drawing.pdf	2	yes	drawing-note text parsing

2.2 Method

Anchors and dowels were identified from the drawing notes (embedded text) and, where a vision-capable model was configured, from the drawn shapes. Every extracted value is listed in Section 4 with its source. The applicable Australian Standards were identified from the fastener type, host material and drawing references (Section 3). The adhesive product was recognised by name, its specification obtained from attached documents or the manufacturer's published data, and compared with what the design requires (Section 5). The strength verification (Section 6) follows the AS 5216:2021 bonded-anchor methodology (EN 1992-4 / EAD 330499 basis) for threaded rods and AS 5216:2021 Appendix D with AS 3600 cl 13.1.2 for post-installed reinforcing bars. Capacity reduction factors are $\phi = 1/\gamma_M$ with the product's installation safety factor γ_{inst} . Calculations are deterministic; language models were used only to read drawings and product text, and never to calculate.

2.3 Limits of this review

- Static ultimate-limit-state loading only. Seismic design (AS 5216:2021 Appendix F), fire, fatigue and impact are not designed.
- Solid concrete, rigid base plate, no torsion; anchor forces from an elastic distribution (plate bearing neglected, conservative).
- Plate bending, welds, bearing and the attached structure are outside the scope (AS 4100).
- AS 5216 is a licensed document: clause numbers shown are those confirmed from public sources; the reviewer should cross-check the factors listed in Appendix A against the licensed copy.

3. Applicable Australian Standards

Primary standard: AS 5216:2021. AS 5216:2021 governs the anchor / dowel design; AS 3600 governs the reinforcement development length and member checks.

Standard	Role	Why it applies	Attached reference
AS 5216:2021 Design of post-installed and cast-in fastenings in concrete	Primary design standard	Chemical (adhesive) anchors / threaded rods are post-installed fastenings in concrete.; Post-installed reinforcing-bar dowels are covered by AS 5216:2021 Appendix D, which refers to AS 3600 for stress development. Relevant: Appendix B (installation, installer competence); Appendix D (post-installed reinforcing bars); Appendix E (redundant non-structural systems); Appendix F (seismic actions) Published 23 July 2021, replaces AS 5216:2018; referenced by NCC 2022. Requires the product to be pre-qualified (assessment to EAD 330499 for bonded anchors; EAD 330087 for post-installed rebar).	not attached
AS 3600:2018 Concrete structures	Supporting - concrete member, reinforcement detailing	Host member is concrete; AS 3600 provides concrete strength, cover, development/lap length (cl 13.1/13.2) and shear-friction (cl 8.4) provisions, and AS 5216 App. D calls it up for post-installed bars. Relevant: cl 13.1 (development of reinforcement); cl 8.4 (shear friction) Edition 2018 with amendments - confirm the current amendment status.	not attached
AS/NZS 1170.0:2002 Structural design actions - General principles	Supporting - design actions	Design loads on the fastening must be ULS factored actions from the AS/NZS 1170 series. Confirm current amendments (Amdt 5:2011 was the latest known to the author).	not attached
AS/NZS 4671:2019 Steel for the reinforcement of concrete	Supporting - reinforcing steel	Reinforcing-bar strength ($f_{sy} = 500$ MPa for 500N/L/E).	not attached

Standard	Role	Why it applies	Attached reference
AS 4291.1 (current edition) Mechanical properties of fasteners - Bolts, screws and studs	Supporting - threaded rod / stud property class	Steel grade of the threaded rod determines f_{yk} and f_{uk} used in steel-failure checks.	not attached
AS 4100:2020 Steel structures	Supporting - attached steelwork	A base plate / bracket is shown; plate bending, bearing and weld design are to AS 4100 (outside this tool's scope). Confirm the current amendment status.	not attached
AS/NZS 4680 (current edition) Hot-dip galvanized (zinc) coatings on fabricated ferrous articles	Supporting - corrosion protection	Galvanised steel is specified.	not attached

Standards cited on the drawings: AS 5216

4. Design data extracted from the drawings

Values are shown with their source (drawing text, vision reading, user entry or assumption). Assumptions are conservative defaults and are repeated as review comments.

4.A1 M20 chemical anchor (p.1)

Parameter	Value	Source
Fastener	Threaded rod M20, grade 8.8	sample_structural_drawing.pdf p.1 - "GRADE 8.8"
Effective embedment h_{ef}	200 mm	sample_structural_drawing.pdf p.1 - "200 EMBED"
Drill hole diameter d_0	22 mm	sample_structural_drawing.pdf p.1 - "22 mm DIA HOLE"
Layout	2 x 2 anchors @ 200/200 mm	drawing / assumed
Edge distances	+x: 150 mm	sample_structural_drawing.pdf p.1 - "EDGE DISTANCE 150"
Concrete	$f'_c = 40$ MPa, cracked, thickness 400 mm	sample_structural_drawing.pdf p.1 - " $f'_c = 40$ MPa"
Adhesive	HIT-RE 500 V3	sample_structural_drawing.pdf p.1 - "cal anchors to be ramset chemset reo 502 xtrem..."
Installation	hammer drilling, dry/wet hole	drawing / assumed
Design action (Drawing)	$N^* = 60$ kN, $V^* = 20$ kN acting in +x (group, ULS)	drawing p.1 - "5. DESIGN LOADS (ULS, GROUP TOTAL): TENSION N^* ..."

4.R1 N16 dowel (p.2)

Parameter	Value	Source
Fastener	Post-installed bar N16, grade 500N	sample_structural_drawing.pdf p.2 - "N16 @ 300 CTS STARTER BARS"
Effective embedment h_{ef}	400 mm	sample_structural_drawing.pdf p.2 - "400 EMBED"
Drill hole diameter d_0	20 mm	sample_structural_drawing.pdf p.2 - "20 mm DIA HOLE"
Bar spacing	300 mm	drawing
Edge distances	remote (not stated)	default / derived
Concrete	$f'_c = 32$ MPa, cracked, thickness 600 mm	sample_structural_drawing.pdf p.2 - " $f'_c = 32$ MPa"
Adhesive	HIT-RE 500 V3	sample_structural_drawing.pdf p.2 - "bars epoxy grouted hilti hit re 500 v3 general..."
Installation	hammer drilling, dry/wet hole	drawing / assumed
Design action (Drawing)	$N^* = 13.5$ kN, $V^* = 0$ kN (per anchor, ULS)	drawing p.2 - "3. DESIGN TENSION IN DOWELS: 45 kN/m ULS."

Information not found on the drawings (requests for information)

- M20 chemical anchor (p.1): direction of the shear force is not stated - assumed to act towards the nearest free edge (conservative). State the shear direction and whether it can reverse.
- N16 dowel (p.2): reinforcement grade not stated - D500N assumed (AS/NZS 4671).

5. Adhesive product assessment

5.1 HIT-RE 500 V3

Source of data	attached documents
Chemistry	not stated
Approvals found	ETA-16/0143, EAD 330499
Seismic categories	C1, C2
Installation safety factor γ_{inst}	hammer: 1, diamond: 1.4, flooded: 1.4
Sustained-load factor ψ_{0sus}	40/24: 0.88, 70/43: 0.7
Data status	machine-read - NOT yet confirmed

Source documents

- ETA-16-0143_excerpt.pdf

Bond resistance for rod $d = 20$ mm (hammer drilled, dry/wet, 50 years)

Temperature range	$\tau_{Rk,cr}$ (MPa)	$\tau_{Rk,ucr}$ (MPa)	Source
I: 40°C/24°C	9.50	16.0	ETA-16-0143_excerpt.pdf p.4
II: 70°C/43°C	7.50	12.0	ETA-16-0143_excerpt.pdf p.4

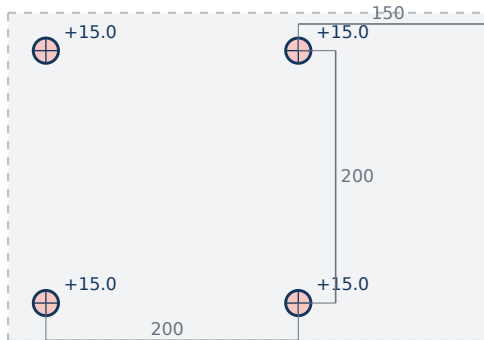
Comparison of design requirements with product specification

Requirement	Needed by the design	Product documents provide	Status
Pre-qualification (AS 5216)	Assessment / ETA to EAD 330499 (bonded anchors) - AS 5216:2021 pre-qualification	ETA-16/0143, EAD 330499 - AS 5216:2021 requires the product to be pre-qualified; installers per Appendix B.	OK
Size coverage ($d = 20$ mm rod)	Bond data for this diameter	yes	OK
Cracked concrete	Cracked-concrete bond resistance $\tau_{Rk,cr}$	present - Design assumes cracked concrete (conservative).	OK
Hole condition (dry/wet)	Data for dry/wet holes	present	OK
Drilling method (hammer)	Qualified for the specified method	present	OK
Embedment range	$h_{ef} = 200$ mm	90 - 400 mm	OK
Installation safety factor	γ_{inst} from the assessment	present	OK
Concrete-strength factor	ψ_c for $f'c = 40$ MPa	present	OK
Data status	Values confirmed against the source by the engineer	machine-read, NOT yet confirmed - Confirm the extracted values against the source document before issuing.	UNKNOWN

6. Structural checks

6.A1 M20 chemical anchor (p.1)

PLAN (mm) - anchor tension kN shown at anchors



solid = free edge; dashed = remote / not stated

Steel - tension	0.11	1.0 limit
Bond / pull-out - group	0.55	
Bond / pull-out - single anchor	0.22	
Concrete cone - group	0.53	
Steel - shear	0.06	
Concrete pry-out	0.09	
Edge failure - +x	0.55	
Interaction N+V - steel	0.02	
Interaction N+V - concrete	0.81	

Result: **PASS** - maximum utilisation 0.81 (Combined tension + shear - concrete failure modes [Drawing])

Assumptions

- Concrete treated as CRACKED (conservative).
- Static ULS loads, rigid base plate, anchors only resist tension (plate bearing ignored), shear shared equally by anchors with standard clearance holes.
- Temperature range not stated: lowest bond resistance (range 'II': 70°C/43°C) adopted.

Load combination	N* (kN)	V* (kN)	Max util.	Governing check	Status
Drawing	60.0	20.0	0.81	Combined tension + shear - concrete failure modes	PASS

Design basis for this case

Symbol	Description	Value
ϕ_c	Concrete-related capacity reduction factor ($1/(\gamma_c \gamma_{inst})$)	0.667
γ_c	Partial factor	1.50
γ_{inst}	Installation safety factor	1.00
$\phi_{Ms,N}$	Steel tension	0.667
$\phi_{Ms,V}$	Steel shear	0.8
k	Cone factor	7.70
ψ_c	Concrete-strength factor on bond	1.07

Anchor forces - governing combination 'Drawing' (kN, + = tension)

Anchor	x (mm)	y (mm)	N (kN)
1	-100.0	-100.0	15.0
2	100.0	-100.0	15.0
3	-100.0	100.0	15.0
4	100.0	100.0	15.0

Calculations - governing combination 'Drawing'

Steel failure - tension (most loaded anchor) EN 1992-4 7.2.1.2 / AS 5216:2021 - steel failure			
A_s	Stress area	245 mm ²	
f_{uk} / f_{yk}	Steel grade 8.8	800 / 640 MPa	
$N_{Rk,s}$	Characteristic steel resistance	196 kN	$A_s \cdot f_{uk}$
γ_{Ms}	Partial factor	1.50	$\max(1.2 f_{uk}/f_{yk}, 1.4)$
$R_k = 196$ kN, $\phi = 0.667$, $R_d = 131$ kN; $E_d = 15.0$ kN; utilisation = 0.11			PASS

Combined pull-out and concrete failure - tension group EN 1992-4 7.2.1 / AS 5216:2021 - bonded anchors, combined pull-out & concrete failure			
τ_{Rk}	Characteristic bond resistance (cracked, dry _{wet}), incl. $\psi_c=1.07$	8.03 MPa	
$N_{Rk,p}^0$	Single-anchor resistance	101 kN	$\pi \cdot d \cdot h_{ef} \cdot \tau_{Rk}$
$s_{cr,Np}$	Critical spacing	506 mm	$7.3 d \sqrt{(\tau_{Rk,ucr})} \leq 3 h_{ef}$
$c_{cr,Np}$	Critical edge distance	253 mm	$s_{cr,Np} / 2$
$A_{p,N} / A_{p,N}^0$	Projected-area ratio	1.66	$A = 603 \times 706 \text{ mm}; A_0 = s_{cr,Np}^2$
$\psi_{g,Np}$	Group factor (n = 4)	1.11	$\psi_0 - (s/s_{cr})^{0.5} (\psi_0 - 1) \geq 1$
$\psi_{s,Np}$	Edge factor	0.878	$0.7 + 0.3 c/c_{cr,Np} \leq 1$
$\psi_{ec,Np}$	Eccentricity factor	1.00	$1/(1+2e/s_{cr})$
$N_{Rk,p}$	Characteristic resistance	164 kN	$N_0 \cdot (A/A_0) \cdot \psi_g \cdot \psi_s \cdot \psi_{ec}$
φ	Capacity reduction factor $1/(\gamma_c \cdot \gamma_{inst})$	0.667	
$R_k = 164 \text{ kN}, \varphi = 0.667, R_d = 110 \text{ kN}; E_d = 60.0 \text{ kN}; \text{utilisation} = 0.55$			PASS

Combined pull-out and concrete failure - most loaded anchor EN 1992-4 7.2.1 / AS 5216:2021 - bonded anchors, combined pull-out & concrete failure			
τ_{Rk}	Characteristic bond resistance (cracked, dry _{wet}), incl. $\psi_c=1.07$	8.03 MPa	
$N_{Rk,p}^0$	Single-anchor resistance	101 kN	$\pi \cdot d \cdot h_{ef} \cdot \tau_{Rk}$
$s_{cr,Np}$	Critical spacing	506 mm	$7.3 d \sqrt{(\tau_{Rk,ucr})} \leq 3 h_{ef}$
$c_{cr,Np}$	Critical edge distance	253 mm	$s_{cr,Np} / 2$
$A_{p,N} / A_{p,N}^0$	Projected-area ratio	1.00	$A = 506 \times 506 \text{ mm}; A_0 = s_{cr,Np}^2$
$\psi_{g,Np}$	Group factor (n = 1)	1.00	$\psi_0 - (s/s_{cr})^{0.5} (\psi_0 - 1) \geq 1$
$\psi_{s,Np}$	Edge factor	1.00	$0.7 + 0.3 c/c_{cr,Np} \leq 1$
$\psi_{ec,Np}$	Eccentricity factor	1.00	$1/(1+2e/s_{cr})$
$N_{Rk,p}$	Characteristic resistance	101 kN	$N_0 \cdot (A/A_0) \cdot \psi_g \cdot \psi_s \cdot \psi_{ec}$
φ	Capacity reduction factor $1/(\gamma_c \cdot \gamma_{inst})$	0.667	
$R_k = 101 \text{ kN}, \varphi = 0.667, R_d = 67.2 \text{ kN}; E_d = 15.0 \text{ kN}; \text{utilisation} = 0.22$			PASS

Concrete cone failure - tension group EN 1992-4 7.2.1.4 / AS 5216:2021 - concrete cone failure			
k	Cone factor (cracked)	7.70	
h_{ef}'	Effective embedment used	200 mm	
$N_{Rk,c}^0$	Single-anchor cone resistance	138 kN	$k \cdot \sqrt{f_c} \cdot h_{ef}^{1.5}$
$s_{cr,N} / c_{cr,N}$	Critical spacing / edge distance	600 / 300 mm	$3 h_{ef} / 1.5 h_{ef}$
$A_{c,N} / A_{c,N}^0$	Projected-area ratio	1.44	$A = 650 \times 800 \text{ mm}; A_0 = 9 h_{ef}^2$
$\psi_{s,N}$	Edge factor	0.85	$0.7 + 0.3 c/c_{cr,N} \leq 1$
$\psi_{re,N}$	Shell-spalling (dense reinforcement) factor	1.00	$0.5 + h_{ef}/200 \leq 1$ (only if reinforcement spacing < 150 mm)
$\psi_{ec,N}$	Eccentricity factor	1.00	$1/(1+2e/s_{cr})$
$N_{Rk,c}$	Characteristic resistance	169 kN	$N_0 \cdot (A/A_0) \cdot \psi_s \cdot \psi_{re} \cdot \psi_{ec}$
φ	Capacity reduction factor $1/(\gamma_c \cdot \gamma_{inst})$	0.667	
$R_k = 169 \text{ kN}, \varphi = 0.667, R_d = 113 \text{ kN}; E_d = 60.0 \text{ kN}; \text{utilisation} = 0.53$			PASS

Steel failure - shear (most loaded anchor) EN 1992-4 7.2.2.3 / AS 5216:2021 - steel failure in shear			
$V_{Rk,s}^0$	Single-anchor steel shear	98.0 kN	$k_6 A_s f_{uk}, k_6 = 0.5$
k7	Ductility factor	1.00	
$\gamma_{Ms,V}$	Partial factor	1.25	$f_{uk}/f_{yk} \geq 1.25$
$R_k = 98.0 \text{ kN}, \varphi = 0.8, R_d = 78.4 \text{ kN}; E_d = 5.00 \text{ kN}; \text{utilisation} = 0.06$			PASS

Concrete pry-out failure EN 1992-4 7.2.2.4 / AS 5216:2021 - concrete pry-out			
k8	Pry-out factor ($h_{ef} = 200 \text{ mm}$)	2.00	
N_{Rk}	Group tension resistance (all anchors; min of bond and cone)	164 kN	
$V_{Rk,cp}$	Characteristic pry-out resistance	328 kN	$k_8 \cdot N_{Rk}$
$R_k = 329 \text{ kN}, \varphi = 0.667, R_d = 219 \text{ kN}; E_d = 20.0 \text{ kN}; \text{utilisation} = 0.09$			PASS

Concrete edge failure - +x edge (c1 = 150 mm) EN 1992-4 7.2.2.5 / AS 5216:2021 - concrete edge failure			
k ₉	Edge-shear factor (cracked)	1.70	
V ^o _{Rk,c}	Single-anchor edge resistance	37.8 kN	$k_9 d^a l f^b \sqrt{f_c} c_1^{1.5}$; $a=0.1(lf/c_1)^{0.5}$, $b=0.1(d/c_1)^{0.2}$
A _{c,V} / A ^o _{c,V}	Projected-area ratio	1.44	$b = 650 \text{ mm}$, $h' = 225 \text{ mm}$; $A_0 = 4.5 c_1^2$
ψ _{s,V}	Corner / side-edge factor	1.00	$0.7 + 0.3 c_2/(1.5 c_1) \leq 1$
ψ _{h,V}	Member-thickness factor	1.00	$(1.5 c_1/h)^{0.5} \geq 1$
ψ _{α,V}	Load-direction factor (α = 0 deg)	1.00	
V _{Rk,c}	Characteristic resistance	54.6 kN	$V_0 \cdot (A/A_0) \cdot \psi_s \cdot \psi_h \cdot \psi_\alpha (\psi_{re,V} = 1.0: \text{no credit for edge reinforcement})$
R_k = 54.6 kN, φ = 0.667, R_d = 36.4 kN; E_d = 20.0 kN; utilisation = 0.55			PASS
• Front row of anchors assumed to carry the full shear (conservative).			

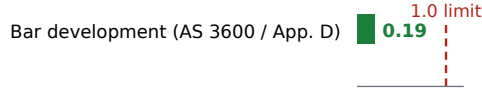
Combined tension + shear - steel EN 1992-4 7.3 / AS 5216:2021			
β _N	N _{Ed} / N _{Rd,s}	0.115	
β _V	V _{Ed} / V _{Rd,s}	0.064	
sum	β _N ² + β _V ²	0.017	≤ 1.0
utilisation = 0.02 (limit 1.00)			PASS

Combined tension + shear - concrete failure modes EN 1992-4 7.3 / AS 5216:2021			
β _N	max N _{Ed} /N _{Rd} (bond, cone)	0.548	
β _V	max V _{Ed} /V _{Rd} (edge, pry-out)	0.55	
sum	β _N ^{1.5} + β _V ^{1.5}	0.813	≤ 1.0
lin	Simplified alternative β _N + β _V	1.10	≤ 1.2
utilisation = 0.81 (limit 1.00)			PASS

Detailing and installation checks

Check	Values	Status
Embedment depth within qualified range	Embedment on drawing: 200 mm; Range (product value if known, else EN default): 90 - 400 mm	PASS
Minimum anchor spacing	Actual minimum spacing: 200 mm; Required minimum (product): 90.0 mm	PASS
Minimum edge distance	Actual minimum edge distance: 150 mm; Required minimum (product): 55.0 mm	PASS
Minimum member thickness	Member thickness: 400 mm; Required (h _{ef} + 2 d ₀): 244 mm	PASS

6.R1 N16 dowel (p.2)



Result: **INCOMPLETE** - maximum utilisation 0.19 (Bar tension capacity - development length (AS 3600 13.1.2 / AS 5216 App. D) [Drawing])

Assumptions

- Static ULS tension in the bar; AS 5216:2021 Appendix D.4.1 development-length method (AS 3600 cl 13.1.2.2) with the adhesive's design bond strength f_{bd} .

Warnings

- △ No f_{bd} (EAD 330087) value found for HIT-RE 500 V3: assumed $\geq$ the EAD table value (3.2 MPa). Confirm against the product's ETA-Rebar.
- △ HIT-RE 500 V3 is not confirmed as qualified for post-installed rebar (EAD 330087 / TR 023) - Appendix D design cannot be relied on until confirmed.
- △ Status downgraded to INCOMPLETE: adhesive qualification for post-installed rebar (EAD 330087) not confirmed.

Load combination	N* (kN)	V* (kN)	Max util.	Governing check	Status
Drawing	13.5	0	0.19	Bar tension capacity - development length (AS 3600 13.1.2 / ...)	PASS

Design basis for this case

Symbol	Description	Value
d_b / f_{sy}	Bar 500N	16 / 500
c_d	min(cover, half clear spacing)	142
k_1, k_2, k_3	AS 3600 13.1.2.2 factors ($k_2 = (132 - d)/100$; $k_3 = 1 - 0.15(c_{d-d})/d$ in [0.7, 1.0])	1, 1.16, 0.70
$L_{sy,t}$ (basic)	$0.5 k_1 k_3 f_{sy} d / (k_2 \sqrt{f'c})$	427
$L_{sy,t}$ (min)	$0.058 f_{sy} k_1 d$	464
$L_{sy,t}$	Development length to yield (governing)	464
$f_{bd,EAD}$	EAD 330087 table design bond strength	3.20
$f_{bd,ETA}$	Product design bond strength (ETA-Rebar)	n/a
L_{req}	Required length = $L_{sy,t} \times \max(1, f_{bd,EAD}/f_{bd,ETA})$	464
L	Embedment provided	400
σ_{st}	Stress developed in bar = $f_{sy} \times \min(1, L/L_{req})$ (AS 3600 13.1.2.4: $L_{st} = L_{sy,t} \sigma_{st}/f_{sy} \geq 12 d_b$)	431
N_{st}	Nominal tensile capacity $A_s \sigma_{st}$	86.7
ϕ	Capacity reduction factor (tension)	0.8

Calculations - governing combination 'Drawing'

Bar tension capacity - development length (AS 3600 13.1.2 / AS 5216 App. D) AS 5216:2021 Appendix D.4.1; AS 3600:2018 cl 13.1.2			
d_b / f_{sy}	Bar 500N	16 / 500 mm / MPa	
c_d	min(cover, half clear spacing)	142 mm	
k_1, k_2, k_3	AS 3600 13.1.2.2 factors	1, 1.16, 0.70	$k_2=(132-d)/100$; $k_3=1-0.15(c_{d-d})/d$ in [0.7,1.0]
$L_{sy,t}$ (basic)	$0.5 k_1 k_3 f_{sy} d / (k_2 \sqrt{f'c})$	427 mm	
$L_{sy,t}$ (min)	$0.058 f_{sy} k_1 d$	464 mm	
$L_{sy,t}$	Development length to yield (governing)	464 mm	
$f_{bd,EAD}$	EAD 330087 table design bond strength	3.20 MPa	
$f_{bd,ETA}$	Product design bond strength (ETA-Rebar)	n/a MPa	
L_{req}	Required length = $L_{sy,t} \times \max(1, f_{bd,EAD}/f_{bd,ETA})$	464 mm	
L	Embedment provided	400 mm	
σ_{st}	Stress developed in bar = $f_{sy} \times \min(1, L/L_{req})$	431 MPa	AS 3600 13.1.2.4: $L_{st} = L_{sy,t} \sigma_{st}/f_{sy} \geq 12 d_b$
N_{st}	Nominal tensile capacity $A_s \sigma_{st}$	86.7 kN	
ϕ	Capacity reduction factor (tension)	0.8	
$R_k = 86.7 \text{ kN}$, $\phi = 0.8$, $R_d = 69.3 \text{ kN}$; $E_d = 13.5 \text{ kN}$; utilisation = 0.19			PASS
• Partial development: bar embedded 400 mm < L_{req} 464 mm - stress limited to 431 MPa; bar cannot reach yield.			

Detailing and installation checks

Check	Values	Status
Minimum concrete cover to post-installed bar (EAD 330087)	Cover not stated; at least 54 mm required for hammer drilling.	WARN
Minimum clear spacing between post-installed bars	Clear spacing: 284 mm; max(40, 4 d): 64.0 mm	PASS
Member thickness vs hole depth	Member thickness: 600 mm; Embedment + minimum rear cover: 454 mm	PASS

Cross-check: bar designed as a bonded anchor (AS 5216 / EAD 330499 basis)

Status FAIL; maximum utilisation 0.17 (Steel failure - tension (most loaded anchor) [Drawing]). Either method may be used by the designer; the Appendix D method is reported as the primary result.

7. Review comments

Critical: 0 Major: 4 Minor: 2 Info: 3

#	Severity	Case	Category	Comment	Action
1	Major	-	Drawing	Information not on the drawing (RFI) M20 chemical anchor (p.1): direction of the shear force is not stated - assumed to act towards the nearest free edge (conservative). State the shear direction and whether it can reverse.	Issue an RFI to the designer.
2	Major	-	Drawing	Information not on the drawing (RFI) N16 dowel (p.2): reinforcement grade not stated - D500N assumed (AS/NZS 4671).	Issue an RFI to the designer.
3	Major	R1	Product	R1: adequacy cannot be confirmed No f_{bd} (EAD 330087) value found for HIT-RE 500 V3: assumed $\geq$ the EAD table value (3.2 MPa). Confirm against the product's ETA-Rebar. HIT-RE 500 V3 is not confirmed as qualified for post-installed rebar (EAD 330087 / TR 023) - Appendix D design cannot be relied on until confirmed.	Attach the adhesive TDS/ETA, run the web lookup, or enter the product values; provide the missing design inputs.
4	Major	R1	Product	R1: product gap - Post-installed rebar qualification Required: EAD 330087 / ETA-Rebar with f_{bd} . Product documents: not confirmed. Rebar-as-anchor design (EAD 330499) is an alternative if bond data exist.	Choose a product that is qualified for this use or obtain the missing data from the manufacturer.
5	Minor	-	Drawing	Drawing note missing No note found for installation temperature and cure time before loading.	Add the note or confirm it is covered in the specification.
6	Minor	R1	Detailing	R1: Minimum concrete cover to post-installed bar (EAD 330087) Cover not stated; at least 54 mm required for hammer drilling.	Confirm.
7	Info	A1	Assumption	A1: assumptions adopted in the calculation Concrete treated as CRACKED (conservative). • Static ULS loads, rigid base plate, anchors only resist tension (plate bearing ignored), shear shared equally by anchors with standard clearance holes. • Temperature range not stated: lowest bond resistance (range 'II: 70°C/43°C') adopted.	Confirm or correct on the drawing / by RFI.
8	Info	R1	Assumption	R1: assumptions adopted in the calculation Static ULS tension in the bar; AS 5216:2021 Appendix D.4.1 development-length method (AS 3600 cl 13.1.2.2) with the adhesive's design bond strength f_{bd} .	Confirm or correct on the drawing / by RFI.
9	Info	R1	Strength	R1: cross-check by anchor method Treating the bar as an adhesive anchor (AS 5216 / EAD 330499 basis) gives max utilisation 0.17 (FAIL). Strength utilisation is within limits but the anchor-method range/detailing rule fails: Embedment depth within qualified range (e.g. embedment beyond the anchor-method range - the Appendix D method applies).	

Appendix A - Design factors used and their verification status

$\phi = 1/\gamma$ for concrete-related failure modes with $\gamma = \gamma_c \cdot \gamma_{inst}$. 'Cross-checked' means the factor reproduces published AS 5216-based manufacturer design tables (see tests in the software). Other factors follow the EN 1992-4 / EAD 330499 methodology on which AS 5216:2021 is based and must be confirmed by the reviewer against the licensed standard.

Symbol	Value	Description	Basis	Cross-checked
k_{cr}	7.70	Concrete cone factor, cracked concrete	EN 1992-4 / EAD 330499 methodology adopted by AS 5216:2021 - Reproduces Ramset SARB cone tables (cracked = 0.70 x uncracked)	yes
k_{ucr}	11.0	Concrete cone factor, uncracked concrete	as above - $11.0 \cdot \sqrt{(32) \cdot 70^{1.5}} / 1.5 = 24.3 \text{ kN} =$ Ramset SARB Table 2a at $h_{ef}=70$	yes
$k9_{ucr}$	2.40	Edge-shear factor, uncracked	as above - Reproduces Ramset SARB Table 4a-1 (M10/M12/M16 at minimum edge)	yes
$k9_{cr}$	1.70	Edge-shear factor, cracked (=0.7 x uncracked)	as above	yes
$k8$	2.00	Pry-out factor ($h_{ef} \geq 60 \text{ mm}$); 1.0 if $h_{ef} < 60 \text{ mm}$	as above - Reproduces Ramset SARB Table 4e (M10 $h=90$: 70.8 kN)	yes
γ_c	1.50	Partial factor, concrete-related failure modes	$\phi = 1/(\gamma_c \cdot \gamma_{inst})$; Ramset SARB uses $\phi = 1/1.5 = 0.67$	yes
γ_{inst}	1.40	Installation safety factor - DEFAULT used only when the product assessment gives none	Conservative upper bound of EAD 330499 categories (1.0/1.2/1.4) - Replace with the value in the product's ETA/assessment	NO - confirm
$\phi_{Ms,N}$	0.667	Steel tension, grade 5.8/8.8 ($\gamma_{Ms} = 1.2$, $f_{uk}/f_{yk} \geq 1.4$)	EN 1992-4; Ramset SARB uses $\gamma_{Ms} = 1.5$ (5.8/8.8), 1.87 (A4-70)	yes
$\phi_{Ms,V}$	0.8	Steel shear, 5.8/8.8 ($\gamma_{Ms,V} = f_{uk}/f_{yk} \geq 1.25$)	EN 1992-4 - Ramset tabulates 0.67 for shear; this tool uses the EN value - confirm vs AS 5216	NO - confirm
ϕ_{rebar}	0.8	Reinforcement in axial tension	AS 3600; used in Ramset SARB worked example ($N^* \leq 0.8 f_{sy} A_b$)	yes
α_{N+V}	1.50	Interaction exponent concrete modes (steel: 2.0); linear alternative 1.2	EN 1992-4 7.3 / Ramset SARB ' $N^*/\phi_{Nur} + V^*/\phi_{Vur} \leq 1.2$ '	yes

Appendix B - Notes read from the drawings

- p.1: DRAWING TITLE: COLUMN BASE PLATE - CHEMICAL ANCHORS
 - p.1: 200 EMBED
 - p.1: SECTION A-A - ANCHOR RODS IN 400 THK CONCRETE SLAB
 - p.1: 1. CHEMICAL ANCHORS TO BE RAMSET CHEMSET REO 502 XTREM OR APPROVED EQUIVALENT.
 - p.1: 2. ANCHORS: 4 No. M20 GRADE 8.8 GALVANISED THREADED ROD, 200 mm MIN EMBEDMENT, 22 mm DIA HOLE.
 - p.1: 4. EDGE DISTANCE 150 mm MIN. ANCHOR SPACING 200 mm.
 - p.1: 7. SCAN FOR EXISTING REINFORCEMENT BEFORE DRILLING. ADHESIVE ANCHORS TO AS 5216.
 - p.2: DRAWING TITLE: WALL STARTER BARS - POST-INSTALLED DOWELS
 - p.2: 400 EMBED
 - p.2: N16 @ 300 CTS STARTER BARS
 - p.2: EPOXY GROUTED - HILTI HIT-RE 500 V3
 - p.2: 1. N16 DOWELS @ 300 CTS EMBEDDED 400 mm INTO EXISTING SLAB WITH HILTI HIT-RE 500 V3 INJECTION EPOXY.
 - p.2: 3. DESIGN TENSION IN DOWELS: 45 kN/m ULS.
 - p.2: 5. INSTALLATION TO BE CARRIED OUT BY AEFAC CERTIFIED INSTALLERS. PROOF LOAD TEST 5% OF DOWELS.
- Standards cited: AS 5216

Appendix C - Reference extracts from attached documents

Attached document	Type	Pages	Standard codes detected
ETA-16-0143 excerpt.pdf	eta	7	-

Appendix D - Limitations and engineer verification

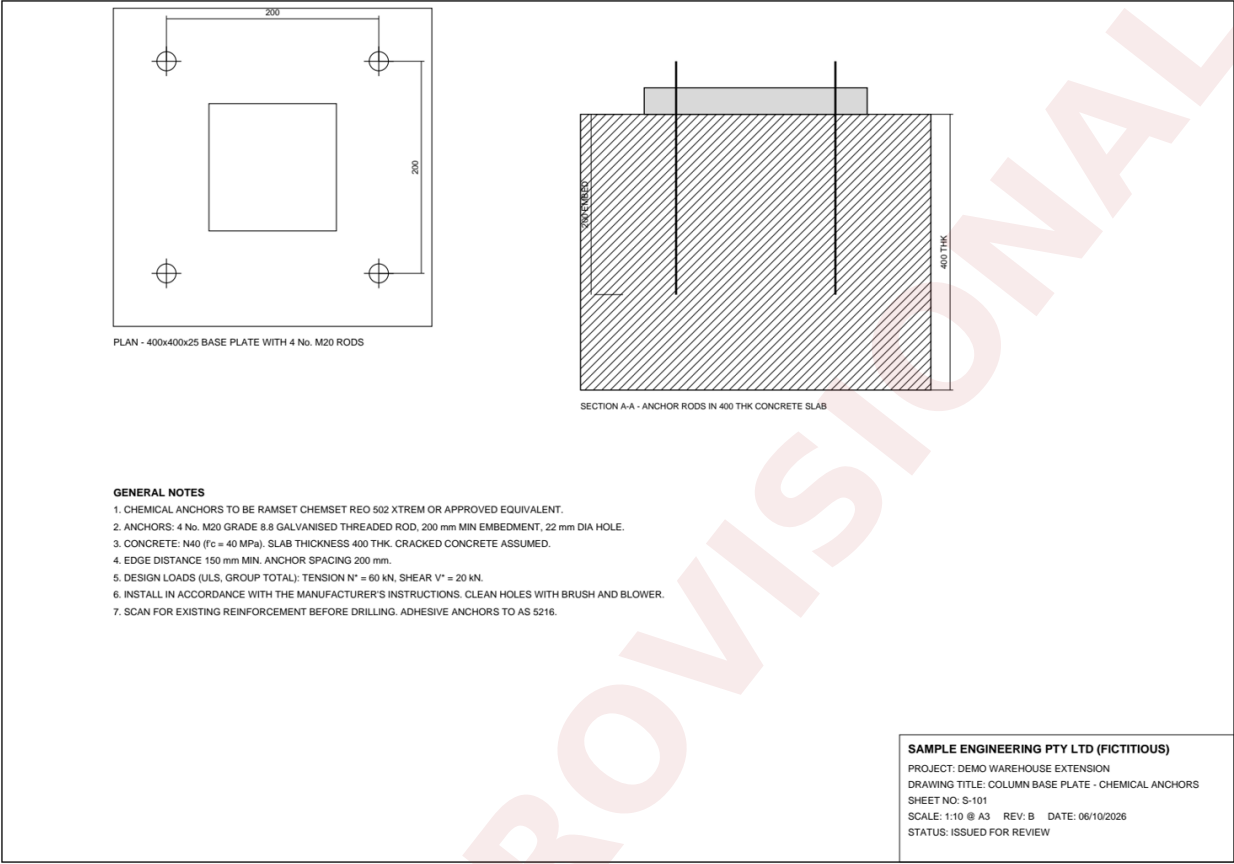
- Product properties marked 'machine-read' were extracted automatically from the quoted source line/page and must be confirmed against the original manufacturer document before issue.
- Where a value is an assumption, the report says so; assumptions are conservative but do not replace project information.

- This report does not constitute certification. Responsibility for the design remains with the registered/chartered structural engineer.
- Installation must follow the manufacturer's instructions and AS 5216:2021 Appendix B (installer competence, hole preparation, proof testing).

Prepared (software-assisted)	Reviewed / verified by engineer
AnchorCheck v0.1.0 06 October 2026	Name: _____ Registration (RPEQ / CPEng / NER): _____ Signature: _____ Date: _____

Appendix E - Drawing pages reviewed

sample_structural_drawing.pdf:1



sample_structural_drawing.pdf:2

